

Conventional Counterspace Conduct and the Outer Space Treaty

Article VI Supervision, Responsible Behaviour and Nationally Anchored Market Conditionality

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Abstract

Article IV of the 1967 Outer Space Treaty does not comprehensively prohibit conventional weapons, anti-satellite capabilities or all military activity in outer space. That proposition does not create a legal vacuum. Conduct not specifically prohibited by Article IV remains governed by the United Nations Charter, the other provisions of the Treaty, applicable rules of international law, registration and liability regimes, and—during armed conflict—applicable international humanitarian law. The legally accurate description is therefore **under-specific regulation**, not *unregulated space warfare*.

This article reassesses a proposal to use Article VI authorisation and continuing supervision as the anchor for nationally implemented restrictions against irresponsible conventional counterspace conduct. Article VI is important but narrower than the original proposal suggested. It imposes international responsibility for national space activities and requires authorisation and continuing supervision of non-governmental national activities. It does not itself create prescriptive jurisdiction over unrelated foreign persons, authorise economic sanctions, or require a State to exclude foreign entities from finance, insurance, procurement or launch services. Those effects require distinct domestic statutory powers and a recognised jurisdictional nexus; they also engage due process, human rights, trade law, comity, blocking legislation and the law of countermeasures.

The revised proposal—**nationally anchored market conditionality**—is modular. Licensing conditions govern national operators. Export controls govern specified goods, software, technology and services within established territorial and nationality jurisdiction. Financial sanctions require a sanctions statute and an independently reviewable designation. Public procurement, insurance and government-backed finance use their own legal bases. International coordination supplies

interoperability and legitimacy, but no measure is presented as an obligation derived solely from Article VI.

The recommended predicate is conduct-based rather than capability-based: destructive testing that creates long-lived debris; unnotified or non-consensual harmful interference; conduct creating a serious and foreseeable collision or cascading-debris risk; evasion of registration and supervision; and material assistance knowingly supplied to such conduct. Safeguards include a clear jurisdictional nexus, published criteria, evidence thresholds, notice, reasons, independent review, proportionality, time limits, humanitarian and safety exceptions, delisting, and periodic multilateral assessment. The mechanism is offered as a policy model, not a statement of existing law.

Keywords: Outer Space Treaty; Article VI; counterspace; anti-satellite; jurisdiction; sanctions; export controls; responsible behaviour; due process; WTO law.

1. Introduction

The contemporary law of military activity in space is often described in two equally misleading ways. One account implies that the Outer Space Treaty prohibits weapons in space generally. The other says that conventional counterspace systems are unregulated because Article IV does not name them. The Treaty supports neither extreme.

Article IV contains a specific weapons-of-mass-destruction prohibition for orbit and outer space, and a more restrictive peaceful-purpose regime for the Moon and other celestial bodies. It does not expressly prohibit every conventional weapon or every military support function in Earth orbit. Article III, however, applies international law and the UN Charter to space activities. Article VI allocates international responsibility and requires supervision of non-governmental national activities. Article VII and the Liability Convention address damage. Article VIII preserves jurisdiction and control over registered objects. Article IX requires due regard, addresses harmful contamination and establishes consultation procedures. Registration, debris mitigation and long-term-sustainability instruments add transparency and risk-reduction expectations [1-6].

The practical problem is therefore not complete absence of law but a mismatch between broad general rules and dual-use, fast-changing conduct. A rendezvous system, servicing vehicle, inspection satellite, launch service or high-powered electromagnetic system may be peaceful, protective or hostile depending on purpose, configuration and use. Capability definitions are both over-inclusive and easy to evade. Conduct-based rules are more promising, but the international system has not yet produced a comprehensive binding instrument.

This article asks whether nationally implemented market conditions can reduce harmful counterspace conduct while multilateral negotiations continue. The answer is conditional. National measures can be lawful and influential, but Article VI cannot bear the entire doctrinal weight. The legal basis must be disaggregated measure by measure.

2. Method and scope

The analysis follows the interpretive framework of Articles 31 and 32 of the Vienna Convention on the Law of Treaties: ordinary meaning, context, object and purpose, followed where appropriate by preparatory work [7]. It distinguishes:

- the content of existing international obligations;
- permissible domestic jurisdiction and regulation;
- policy choices available under domestic law; and
- my own normative proposal.

The article is not legal advice and does not assess any current operator, State or incident. “Counterspace conduct” is used functionally to describe conduct intended or foreseeably likely to deny, degrade, disrupt, damage or destroy a space system or its supporting service. The term does not determine legality. No engineering or operational detail is included.

3. Article IV: a limited prohibition within a larger legal order

3.1 The two-tier text

Article IV's first paragraph prohibits placing in orbit around Earth objects carrying nuclear weapons or other weapons of mass destruction, installing such weapons on celestial bodies, or stationing them in outer space in any other manner. Its second paragraph reserves the Moon and other celestial bodies exclusively for peaceful purposes and prohibits military bases, installations and fortifications, weapons testing and military manoeuvres there. It also clarifies that military personnel and equipment may be used for scientific or other peaceful purposes [1].

The textual consequences are:

1. the orbit-wide prohibition is weapon- and payload-specific;
2. the celestial-body regime is activity- and location-specific; and
3. the Treaty does not contain a general, express prohibition on all conventional weapons in Earth orbit.

That conclusion should be stated carefully. It is not a finding that any particular conventional counterspace operation is lawful.

3.2 “Peaceful purposes”

State practice and the structure of Article IV make it difficult to read *peaceful* as *non-military* throughout outer space. Military communication, navigation, warning and observation have long coexisted with the Treaty. The more defensible reading is that military activity is not automatically unlawful, while aggression and other conduct contrary to the Charter remain prohibited. The

interpretation does not eliminate debate about the stronger celestial-body wording or about particular threatening conduct [8-10].

3.3 Article III and the UN Charter

Article III requires activities in outer space to be carried on in accordance with international law, including the Charter, in the interest of maintaining international peace and security. The Charter's prohibition on the threat or use of force therefore applies in space. Whether a particular interference amounts to a prohibited use of force or an armed attack depends on facts, scale, effects, context, attribution and legal justification. The Treaty does not answer those questions by hardware label [11-13].

International humanitarian law applies when the threshold and scope of an armed conflict are met. Its application does not legitimise resort to force; it regulates conduct during conflict. Dual-use space services create difficult questions of distinction, proportionality, precautions and reverberating civilian effects [13,14].

3.4 The accurate baseline

Conventional counterspace systems are **not specifically prohibited as a class by Article IV**, but their development and use are governed by a layered legal order. "Not expressly banned by Article IV" is not equivalent to "lawful", "unregulated" or "free from responsibility".

4. The ancillary regime

4.1 Due regard and consultations

Article IX requires States Parties to conduct activities with due regard to the corresponding interests of all other States Parties. A State that has reason to believe an activity would cause potentially harmful interference must undertake appropriate international consultations before proceeding. The provision is procedurally valuable, but its open texture leaves uncertainty about thresholds, remedies and emergency situations [1,15].

Debris-creating conduct is particularly relevant because collision risk and loss of orbital utility extend beyond the immediate parties. The COPUOS Space Debris Mitigation Guidelines and Guidelines for the Long-term Sustainability of Outer Space Activities are non-binding, but they help specify reasonable risk-reduction practice and foreseeability [5,6].

4.2 Registration and jurisdiction

The Registration Convention requires specified information about launched space objects. Article VIII of the Outer Space Treaty links registration to jurisdiction and control. Registration improves attribution and communication but does not disclose complete capability, payload or manoeuvre

intent. A responsibility mechanism should therefore reward timely, accurate registration without treating registration itself as proof of benign conduct [1,4,16].

4.3 Responsibility and liability

Article VI concerns international responsibility for national activities. Article VII and the Liability Convention concern liability for damage caused by space objects. Responsibility and liability are related but distinct. Liability allocates consequences after damage; it does not by itself determine whether an activity was lawful before damage. Fault-based liability for damage elsewhere than the surface of Earth also leaves evidentiary and causal difficulties [1-3,17].

5. Article VI: indispensable but not jurisdiction-conferring

5.1 What Article VI requires

Article VI provides that States Parties bear international responsibility for national activities in outer space, whether conducted by governmental agencies or non-governmental entities, and for assuring conformity with the Treaty. Activities of non-governmental entities require authorisation and continuing supervision by the appropriate State Party [1].

This is a strong basis for national space-activity licensing. A State must have some means to authorise, monitor and, where necessary, condition or terminate the national non-governmental activities for which it is responsible. Licensing conditions can address registration, collision avoidance, safety, cybersecurity, debris mitigation, financial responsibility and compliance reporting, subject to domestic law.

5.2 What Article VI does not do

Article VI does not:

- define the full scope of *national activity* or identify one exclusive State of supervision in every multinational chain;
- directly confer domestic prescriptive jurisdiction over every foreign company whose conduct affects space;
- authorise asset freezes, payment restrictions or secondary sanctions;
- create export-control powers;
- require private banks, insurers, cloud providers or launch companies to withdraw service from unrelated foreign entities;
- displace domestic constitutional and administrative-law safeguards; or
- excuse inconsistency with other international obligations.

The “appropriate State Party” wording assumes an allocation of regulatory authority; it does not create unlimited authority. A measure aimed at a national operator may be the performance of

Article VI supervision. A measure aimed at a foreign manufacturer with no territorial, nationality, corporate, transactional or effects nexus needs another jurisdictional and statutory basis.

5.3 National activities in multinational supply chains

Modern missions may involve a spacecraft owner in one State, an operator in another, a launch from a third, components and software from many others, and financing or insurance elsewhere. Multiple States may have legitimate bases for regulation. The solution is coordination and allocation, not an assumption that one Article VI licence automatically governs the world.

Licences should identify:

- the regulated legal person and jurisdictional nexus;
- the licensed activity and object;
- supervisory access and reporting;
- material changes requiring approval;
- transfer-of-control rules;
- subcontractor and beneficial-ownership disclosure;
- incident reporting; and
- suspension, revocation and appeal.

6. Current multilateral policy

6.1 Treaty and behaviour-based tracks

The Conference on Disarmament has long considered prevention of an arms race in outer space. The proposed PPWT illustrates the attraction and difficulty of a binding placement-focused instrument: definitions, dual use, ground-based systems, verification and prohibited conduct remain contested [18,19].

The behavioural track asks what States do rather than what an object is. United Nations processes on responsible behaviours, risk reduction and PAROS have developed proposals around harmful interference, close approaches, notifications, debris and transparency. This approach does not remove the need for legal rules, but it is better matched to dual-use technology.

6.2 Destructive direct-ascent testing

UN General Assembly Resolution 77/41 called on States to commit not to conduct destructive direct-ascent anti-satellite missile tests [20]. The resolution and unilateral commitments are not a comprehensive treaty prohibition. They are nevertheless significant evidence of a conduct-focused norm: the objection is the deliberate destructive test and its debris risk, not possession of every technology capable of reaching orbit.

6.3 The 2025-2028 OEWG

General Assembly Decision 79/512 established a single Open-ended Working Group on the Prevention of an Arms Race in Outer Space in all its Aspects, adopted 174-2-9 on 2 December 2024 [21]. As of 23 July 2026 the group's third substantive session had been scheduled for 6-10 July 2026, with further sessions through 2028 [22]. The process is current and relevant but should not be represented as having produced a binding comprehensive settlement.

6.4 The scholarly disagreement this article inherits

The doctrinal baseline of Section 3 is contested at both ends, and the proposal must survive the contest rather than ignore it. On the permissive end, Bourbonnière and Lee's analysis of conventional weapons in orbit concludes that placement is not per se prohibited and that the law of armed conflict would govern use [31]—a position broadly consistent with the baseline adopted here, but reached through an armed-conflict frame this article deliberately subordinates to peacetime supervision. On the restrictive end, arguments from the “peaceful purposes” principle would read Article IV more broadly than its text [33]; Section 3.2 explains why that reading is not sustained by practice. Koplow's customary-law argument that debris-generating ASAT testing is already unlawful [32] runs parallel to the conduct predicate proposed here, and the open-source counterspace record shows the conduct in question is not hypothetical [34]. The manual projects—Woomera [9], MILAMOS [41] and, by structural analogy, Tallinn 2.0 [40]—map the same under-specified terrain; their persistent points of non-consensus are one reason this article prefers nationally anchored measures to declaratory restatement.

The sharper disagreement concerns effectiveness. The sanctions-scepticism literature—from the comprehensive empirical surveys [35] through the “sanctions sometimes smart” reassessment [36], the UN targeted-sanctions data showing coercion succeeding in a minority of episodes [37], and the historical account of sanctions as a blunt economic weapon [38]—warns that measures of this family underperform their political billing, and the secondary-sanctions critique warns that they overreach jurisdictionally when they try to compensate [39]. I accept both warnings and build the proposal around them: the predicate is conduct-based and evidentiary rather than status-based, precisely the design feature the targeted- sanctions literature associates with its *constraining* and *signalling* successes rather than its coercive failures [37]; each measure is confined to an established jurisdictional nexus, which is the specific discipline the secondary-sanctions critique demands [39]; and the objective claimed is norm-hardening through market conditionality in the sense of the weaponised-interdependence and Brussels-effect literatures [23,24], not behavioural coercion of a determined state actor. If the sceptics are right about coercion, the proposal still does its work; if they are wrong, it does more. That asymmetry, not optimism about sanctions, is the basis on which the proposal lands where it does. Registration-practice and precedent anchors—including the Cosmos 954 settlement [43] and the standard treatise treatment [44], alongside the Artemis Accords as a contemporary plurilateral instrument [42]—complete the frame within which Section 7 operates.

7. Revised proposal: nationally anchored market conditionality

7.1 Concept

Nationally anchored market conditionality (“NAMC”) is a coordinated set of domestic measures that conditions access to regulated space activities, controlled technology, public procurement, government-backed finance, insurance or selected financial services on responsible counterspace conduct. Its influence derives from legally regulated market nodes. It is not presented as universal jurisdiction or as a treaty obligation.

The model is:

conduct predicate → jurisdictional nexus → measure-specific legal power

→ procedural safeguards → proportionate consequence

Every arrow must be independently justified.

7.2 Conduct predicates

The preferred predicates are observable and effects-oriented:

1. conducting a destructive counterspace test that creates long-lived orbital debris;
2. intentionally causing harmful interference with a space object without consent or a recognised legal justification;
3. conducting an unnotified operation that creates a serious and foreseeable collision or cascading-debris risk;
4. materially misrepresenting registration, control or mission information to evade lawful supervision;
5. refusing required Article IX consultation in circumstances of potentially harmful interference; or
6. knowingly supplying material assistance specifically directed to conduct in categories 1-5.

The last predicate must use knowledge, materiality and causation thresholds. Ordinary general-purpose supply should not create strict liability.

7.3 Capability neutrality

A capability-based list would capture servicing, inspection, debris removal and planetary-defence technologies that may have legitimate functions. NAMC therefore targets conduct and evasion. Capability may inform risk and licence conditions, but it is not sufficient for punitive designation.

8. Separate legal bases for separate measures

8.1 Space-activity licensing

Licensing is the closest implementation of Article VI. A State may condition authorisation of national non-governmental activity on:

- compliance with the Treaty and applicable international law;
- accurate registration and beneficial-ownership information;
- collision-avoidance and conjunction coordination;
- debris mitigation and disposal;
- incident notification and preservation of evidence;
- continuing technical and financial capability; and
- non-participation in specified irresponsible conduct.

The domestic space statute must define regulated activities, extraterritorial reach, enforcement, emergency powers and appeal.

8.2 Export controls

Export and technology controls require export-control legislation. They may apply to goods, software, technical data, brokering, re-export and services within the statute's territorial, nationality or item-based reach. Controls should identify controlled functions and end-use risks, provide licensing and exceptions, and avoid treating broad scientific knowledge as automatically controlled.

Article VI may explain the policy interest in responsible national activity. It does not itself supply the export prohibition.

8.3 Financial sanctions

Asset freezes, transaction prohibitions and restrictions on financial services require a sanctions statute or other domestic power. A designation should identify the responsible person, conduct, evidentiary basis and nexus. Direct measures against a person within the enacting State's jurisdiction are legally different from "secondary" measures intended to change an unrelated foreign person's dealings with a third party.

Secondary sanctions can produce substantial extraterritorial effects through intermediary risk management. Those effects are not transformed into jurisdiction merely because private actors choose over-compliance. Comity, conflicting foreign law, blocking statutes, rights of defence and diplomatic consequences must be assessed [23-25,39], and the empirical record of sanctions effectiveness counsels modest expectations for any coercive objective [35-38].

8.4 Procurement and government-backed finance

Governments have distinct powers as purchasers, grant-makers and providers of export credit or insurance. They may impose responsible-conduct conditions on eligibility, subject to procurement law, non-discrimination rules, proportionality and applicable trade commitments. A public-contract exclusion is not the same legal act as a financial sanction and should not borrow its procedure by analogy without statutory support.

8.5 Private insurance and finance

States may require licensed operators to carry insurance or demonstrate financial responsibility. They may also regulate insurers and financial institutions within jurisdiction. A policy merely urging private firms to exclude foreign actors creates opacity and over-compliance risk. Mandatory restrictions should be legally defined; voluntary risk policies should not be misrepresented as public adjudications.

8.6 Civil and administrative liability

Domestic law can allocate remediation cost, insurance, penalties and licence consequences for national operators. Such rules should distinguish preventive regulation from international claims under the Liability Convention and from State responsibility.

9. Jurisdiction, due process and proportionality

9.1 Jurisdictional nexus

A legally cautious measure states its basis. Common bases include:

- territory, including conduct or services performed in the State;
- nationality of a regulated person;
- incorporation or establishment;
- control over a licensed object or activity;
- transaction through a regulated financial or service provider;
- protective jurisdiction for a genuine security interest, within accepted limits; and
- effects, where recognised by the enacting legal order and tempered by reasonableness and comity.

NAMC rejects a free-standing claim that “space is global, therefore every State may regulate every actor”.

9.2 Procedural safeguards

The minimum designation procedure should include:

1. published predicates and legal authority;
2. an identifiable evidence standard;
3. a statement of reasons, subject to narrowly tailored security redactions;
4. notice where compatible with asset-preservation needs;
5. a prompt opportunity to respond;
6. independent administrative and judicial review;
7. periodic review and automatic expiry unless renewed;
8. a clear delisting and error-correction route;
9. humanitarian, safety, incident-response and remediation exceptions; and
10. publication of aggregate implementation data.

Secret evidence may sometimes be necessary, but a system that cannot disclose the gist of the case or permit effective challenge risks arbitrary designation and poor error correction.

9.3 Attribution and corporate responsibility

Space conduct may be carried out through State entities, private operators, contractors and layered corporate vehicles. Legal attribution to a State and administrative designation of a company are separate enquiries. A company should not be designated solely because its State of incorporation engaged in unrelated conduct.

Material assistance predicates should require:

- identified underlying conduct;
- a substantial contribution;
- actual knowledge or a defined reason-to-know standard;
- a causal or purposive link; and
- proportional treatment of remediation and withdrawal.

9.4 Proportionality

Consequences should escalate:

- enhanced reporting or licence conditions;
- denial of a specific controlled export;
- suspension from a defined public contract;
- targeted service restrictions;
- time-limited asset restrictions for serious conduct; and
- coordinated measures for repeated or aggravated violations.

A global economic exclusion for a minor reporting failure would be disproportionate. The design should create an incentive to stop, disclose and remediate harmful conduct.

10. Trade law and conflicting obligations

10.1 WTO disciplines

Import, export, procurement and service restrictions may engage GATT, GATS and other WTO obligations. A restriction should identify the *prima facie* rule, possible exception and evidence. GATT Article XX contains closed general exceptions subject to its chapeau; Article XXI contains security exceptions. WTO panels have treated aspects of Article XXI as reviewable rather than entirely self-judging, including in *Russia—Traffic in Transit* [26-29].

The legal result is measure- and fact-specific. “National security” is not a substitute for analysis. A responsible statute should use clear predicates, record the connection to an essential security interest, and consider less-trade-restrictive alternatives where relevant.

10.2 Procurement rules

Government-procurement disciplines vary by membership, coverage schedules and exceptions, so no general answer is available: the same responsible-conduct condition may be unobjectionable in an uncovered procurement and contestable in a covered one. Conditions should be tied to contract performance or a lawful exclusion ground and applied consistently.

10.3 Blocking statutes and comity

Another State may prohibit compliance with extraterritorial sanctions or allow recovery of resulting damage. Operators can face irreconcilable obligations. Coordination among major jurisdictions reduces conflict; it does not eliminate the need for statutory clarity, waivers and conflict-of-laws analysis.

10.4 Countermeasures

State countermeasures are governed by the law of State responsibility and respond to a prior internationally wrongful act. They require attribution, purpose, proportionality and procedural conditions. NAMC is designed primarily as prospective domestic regulation. It should not casually label every restriction a countermeasure in order to avoid treaty or trade analysis [30].

11. Institutional design

11.1 National authority

An implementing State should designate a lead authority but preserve specialised roles:

- space regulator for Article VI licensing;
- export-control authority for controlled items and services;
- sanctions authority for financial restrictions;

- procurement bodies for public-contract eligibility;
- competition and financial regulators for over-compliance and market effects;
- courts for independent review; and
- foreign ministry for consultation and multilateral coordination.

Conflating these functions in one opaque list would reproduce the doctrinal error the revised proposal is intended to correct.

11.2 Multilateral coordination

Participating States could publish a shared non-binding schedule of conduct predicates and evidentiary factors, exchange reasons and incident data, and recognise remediation. Each State would still enact and review measures under its own law. Coordination should be open to revision through the UN processes and should not purport to replace them.

11.3 Independent technical panel

A small technical panel could assess debris persistence, harmful interference, registration evidence and remediation. It should not make binding legal designations. Separating technical findings from legal decisions improves reviewability.

12. Objections and limits

12.1 Fragmentation

Different national lists may fragment the space economy. Shared predicates, published reasons and reciprocal consultation reduce but cannot remove this risk. The answer is not to pretend Article VI automatically harmonises domestic law.

12.2 Politicisation

Any designation regime can be used selectively. Clear conduct predicates, cross-bloc evidence, time limits and independent review are essential. Capability-based or nationality-based presumptions should be avoided.

12.3 Evasion

Actors may change ownership, flags, launch providers or payment routes. Beneficial-ownership and anti-circumvention rules can address deliberate evasion, but broad association liability would chill legitimate commerce.

12.4 Limited leverage

States or entities with captive supply chains may not depend on participating markets. NAMC is therefore a complement to diplomacy, verification, confidence-building, technical resilience and binding law—not a substitute.

12.5 Risk of escalation

Restrictions framed as punishment for security conduct may prompt retaliation. Graduated consequences, consultation, evidence transparency and off-ramps are not procedural luxuries; they are escalation controls.

13. Model legislative principles

A domestic framework should:

1. define the covered national space activity and person;
2. identify each separate licensing, export, financial and procurement power;
3. use conduct-based predicates;
4. require a jurisdictional nexus;
5. define knowledge and material-assistance standards;
6. permit urgent interim measures only with prompt review;
7. require reasons and an effective challenge;
8. apply proportionality and fixed review periods;
9. protect emergency safety, remediation and humanitarian services;
10. coordinate internationally while preserving domestic legal accountability;
11. publish aggregate outcomes and error corrections; and
12. require periodic legislative evaluation.

These principles are deliberately less dramatic than an automatic global blacklist. Their restraint is what makes them legally more credible.

14. Conclusion

The Outer Space Treaty neither comprehensively bans conventional counterspace systems nor leaves them outside law. Article IV is partial. Articles III, VI, VII, VIII and IX, the Charter, registration and liability regimes, applicable humanitarian law and domestic regulation form a layered framework whose general rules often lack the specificity of contemporary dual-use conduct.

Article VI is a sound anchor for authorising and supervising national non-governmental space activity. It is not a charter for worldwide sanctions. The revised proposal therefore disaggregates legal authority. Licences, exports, finance, procurement, insurance and liability each require their own power, nexus and safeguards.

Nationally anchored market conditionality may help translate an emerging conduct-based norm into practical incentives while the 2025-2028 OEWG and other multilateral processes continue. Its legitimacy depends on published predicates, evidence, review, proportionality and coordination. Without those features it would risk becoming the arbitrary extraterritorial coercion it claims to replace.

The proposal is an original normative model. It is not existing international law, and no DOI or preprint status would make it so.

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